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Natural deep eutectic solvents-*platostoma palustre* polysaccharides/polyvinyl alcohol nano-electrospun membranes: Fabrication, characterization, transdermal property and bioactivity

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ABSTRACT

Conventional wet facial masks face critical environmental and functional limitations, including reliance on non-biodegradable polymers, preservative-dependent formulations, and inefficient transdermal delivery of bioactive macromolecules. This study developed eco-friendly nanofibrous facial membranes by integrating natural deep eutectic solvents (NADES) with electrospinning to enhance *Platostoma palustre* polysaccharide (PPP) delivery. The L-proline/lactic acid NADES (L-proLA) increased PPP solubility 51.63-fold through hydrogen-bond disruption, confirmed by FT-IR analysis and Kamlet-Taft parameters ($\alpha = 2.904$). Electrospinning with polyvinyl alcohol (PVA) created 40–150 nm diameter nanofibers (DPNMs) enabling dual transdermal mechanisms: lactic acid's stratum corneum disruption and sustained nanofiber release. DPNM-12 achieved 10.11 ± 0.13 mg/cm^2 cumulative skin permeation in 24 h. The membranes demonstrated 95.36 ± 0.17 % *Staphylococcus aureus* and 93.59 ± 0.15 % *Escherichia coli* inhibition, suppressed macrophage nitric oxide by 75.8 %, and maintained >90 % fibroblast viability. Utilizing food-grade biodegradable components (PVA, NADES) eliminated organic solvents, establishing a green platform for polysaccharide-based skincare. This technology addresses conventional masks limitations by combining antimicrobial/anti-inflammatory functions with scalable eco-friendly production.

1. Introduction

The skin, the body's largest organ covering about 2 m^2, serves as the primary defense against infections and external substances [1]. Skin appearance indicates health or disease, making people with specific skin conditions prone to psychological distress [2]. Therefore, with the gradual improvement in living standards, the demand for beauty and skincare products has witnessed substantial growth over the past few decades. Among them, facial masks have become an integral part of skincare routines for individuals seeking effective solutions for issues such as acne, aging, and hydration. Notably, the global facial masks market reached US $8.54 billion in 2022 and is anticipated to project robust growth in the forecast period with a Compound Annual Growth Rate (CAGR) of 7.2 % through 2028 [3,4]. However, the extensive production of facial masks causes significant environmental pollution, resource waste, and other adverse impacts [5]. These issues drive research on facial masks toward natural, bio-based materials and sustainable technologies for producing eco-friendly facial masks with minimal additives, waste and pollution.

Conventional wet facial masks consist of non-woven fabrics and liquid essence. The nonwoven fabrics are typically made from non-biodegradable petrochemical-based plastics. The high moisture and nutrient content of the liquid essence promotes microbial growth, necessitating non-eco-friendly chemical preservatives, which can also cause skin allergies and inflammation [6]. Electrospinning creates dry facial masks that overcome wet mask limitations by eliminating preservative requirements and minimizing packaging through dry storage [7]. Additionally, electrospun nanomembranes also exhibit notable features such as small fiber diameter, high porosity, large surface area, simple structure, and high drug-loading capacity [8]. Electrospun dry

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facial masks have become a key research focus in sustainable skincare due to their distinct advantages. These nanomembranes integrate active ingredients within synthetic polymer carriers, requiring careful optimization of solubility compatibility for successful co-spinning [9].

Moreover, the cosmetics industry now favors natural, biodegradable ingredients that fit low-carbon lifestyles [10]. Macromolecular polymers are essential raw materials in cosmetics. Among these, natural polysaccharides, a type of carbohydrate polymer, are abundant in nature and can be extracted from plants, algae, animals, and fungi [11]. Recent advances in polysaccharide engineering have demonstrated their potential as multifunctional platforms — particularly through pH-sensitive hydrogel systems that combine controlled delivery with inherent bioactivity [12]. Carboxamide-functionalized cellulose hydrogels developed through the Passerini three-component reaction (P-3CR) exemplify modified polysaccharides with dual benefits. These materials effectively resolve delivery challenges while demonstrating potent antimicrobial activity against common pathogens including *Escherichia coli* (*E. coli*) and *Staphylococcus aureus* (*S. aureus*). Natural polysaccharides possess distinct structural characteristics—including molecular weight, monosaccharide composition, charge properties, and glycosidic linkages—that govern their functional capabilities. These inherent features enable diverse applications across industries ranging from food and medicine to cosmetics [13]. Significantly, in cosmetic formulations, due to their safety, biocompatibility, and diverse biological activities, polysaccharides are superior to other polymers. Multicomponent reaction-based bioactive scaffolds show particular promise in cosmetics. Their pH-responsive carriers enable adaptive release of active ingredients while maintaining intrinsic antibacterial properties [12]. However, owing to the high molecular weight, charge properties and intermolecular/intramolecular hydrogen bonds of polysaccharides, they exhibit poor water solubility and skin permeability hindering practical use [14]. Advancing polysaccharide applications in cosmetics effectively necessitates addressing critical challenges in optimizing their solubility and augmenting their transdermal delivery capabilities.

Natural deep eutectic solvents (NADES), composed of natural materials such as sugars, organic acids, amino acids, and organic compounds, are formed through hydrogen bonding interactions between acceptors (HBAs) and donors (HBDs) [15]. It is the next-generation green solvent derived from ionic liquid (ILs) and deep eutectic solvent (DES), offering similar properties but with improved safety and sustainability [16]. With proven efficacy in enhancing polysaccharide solubility and transdermal delivery, NADES represents a promising green solvent technology. It addresses key challenges and advances the development of sustainable polysaccharide-based cosmetics. Previous studies have established NADES as effective solvents for biopolymers. Huang et al. demonstrated that the choline chloride-malic acid (C-M) system efficiently extracted minerals and proteins from shrimp shells, producing high-quality chitin. Similarly, Kumar et al. reported that a lactic acid, choline chloride, and water mixture dissolved and recovered high-purity lignin and xylan, with the recovered NADES maintaining consistent performance over three recycling cycles [17,18]. However, the solubility of plant polysaccharides in NADES remains underexplored, with limited insights into the dissolution mechanisms of biopolymers. These gaps highlight the need for further investigation. Meanwhile, NADES has exhibited significant potential as penetration enhancers in transdermal drug delivery systems (TDDS). By disrupting the stratum corneum lipid bilayer and forming hydrogen bonds with active ingredients, NADES enhances their permeability through the skin. The properties of NADES, such as natural composition, biocompatibility and low toxicity, make them superior alternatives to conventional enhancers, which often cause irritation and skin damage. Additionally, NADES has shown potential in stabilizing high molecular weight compounds. Tanner et al. reported the impact of ion ratios in choline-geranic acid (CAGE)-based NADES on physicochemical properties and transdermal insulin delivery, revealing that compositions with excess geranic acid significantly enhanced skin permeability and drug transport [19,20].

Platostoma palustre (Blume) A. J. Paton, commonly known as *liangfengcao* or *xiancao*, is an annual herbaceous species within the Lamiaceae family. This plant exhibits a natural distribution pattern across coastal Asia, spanning China, Vietnam, Malaysia, Thailand, and Indonesia. In traditional Chinese medicine systems, it has been historically valued for its therapeutic properties including heat-clearing, detoxification, antioxidant activity, along with demonstrated antihypertensive and hypoglycemic effects [21]. Beyond medicinal applications, the species serves as a culinary resource for preparing characteristic regional products such as herbal jelly and functional teas. Recent phytochemical investigations have identified *Platostoma palustre* polysaccharide (PPP) as its principal bioactive constituent, exhibiting pronounced antioxidative capacity and metabolic regulatory functions in glucose and lipid homeostasis [22]. These pharmacological attributes have facilitated PPP's current industrial applications as a natural gelling agent and nutraceutical additive. Notably, the plant's dual-use paradigm as both medicinal and edible resource positions it as a promising candidate for sustainable product development. Particularly in cosmetic formulations, PPP's multifunctional properties combined with the species' ecological abundance and cost-effectiveness present opportunities for creating green cosmetic ingredients that align with circular economy principles. Such cross-sectoral utilization not only enhances resource efficiency but also expands the value chain of this ethnobotanically significant species.

In this study, we presented a synergistic strategy integrating NADES-enabled polysaccharide engineering with electrospinning technology to develop multifunctional facial mask membranes. Six NADES variants were rationally designed to probe structure-solubility relationships, with L-proline/lactic acid (L-proLA) NADES identified as optimal for PPP dissolution through combined hydrogen bond disruption and hydrophobic interaction mechanisms. Subsequent electrospinning with PVA yielded nanofibrous membranes (DPNMs) exhibiting uniform morphology (fiber diameter: 40 ± 20 nm), high PPP loading efficiency, and controlled release kinetics. Comprehensive biocompatibility assessments and *in vitro/ex vivo* penetration studies utilizing FITC-labeled PPP confirmed DPNM's ability to enhance epidermal permeation compared to free PPP while maintaining over 90 % cell viability in L929 fibroblasts. Antimicrobial assays further revealed 99 % inhibition against *Propionibacterium acnes* (*P. acnes*), underscoring DPNM's potential as a holistic solution for acne-prone skin. By establishing NADES-electrospinning as a platform technology for plant polysaccharide valorization, this work advanced the development of high-performance, eco-conscious skincare products aligned with circular economy principles.

2. Materials and methods

2.1. Drugs and reagents

Sulfuric acid (H_2SO_4), acetic anhydride, chloroform ($CHCl_3$), trichloroacetic acid, sodium carbonate (Na_2CO_3), phenol, calcium chloride ($CaCl_2$), magnesium chloride ($MgCl_2$), DL-malic acid, betaine, L-proline, citric acid, β-alanine, glycerol, and lactic acid (Macklin Biochemical); polyvinyl alcohol (PVA, hydrolysis degree: 87.0–89.0 mol%, viscosity: 80.0–110.5 mPa·s) (Aladdin Biochemical); Mueller-Hinton broth (Thermo Fisher); BCA protein quantification kit, *m*-hydroxyphenyl derivatives (Tixiai Chemical); fetal bovine serum (FBS), phosphate-buffered saline (PBS), streptomycin, penicillin, Dulbecco's Modified Eagle Medium (DMEM; high glucose), and 0.25 % trypsin-EDTA (Gibco); L929 fibroblasts and RAW264.7 macrophages (Chinese Academy of Sciences Cell Bank); CCK-8 assay kit (Solarbio); nitric oxide (NO) detection kit (Biling Bio-Tech); *Staphylococcus aureus* (*S. aureus*), *Escherichia coli* (*E. coli*), *Propionibacterium acnes* (*P. acnes*) (Guangdong Provincial Microorganism Culture Collection Center). All chemicals were used as received without further purification. Deionized water was

used for all experiments.

2.2. Preparation and characterization of PPP

2.2.1. Preparation of PPP

Platostoma palustre (Blume) A. J. Paton was procured from Meizhou, Guangdong Province, China and polysaccharides were extracted through a modified literature-based protocol following these steps [23]. First, air-dried herbs were pulverized and sieved through a 20-mesh sieve, followed by defatting through 95 % ethanol immersion. Subsequently, 200 g of the defatted powder underwent two cycles of ultrasonic-assisted extraction (300 W, 30 min per cycle) using 0.125 M Na_2CO_3 solution at a 1:20 solid-to-liquid (S/L) ratio. The resulting extract was concentrated to one-third volume, after which ethanol precipitation was performed to achieve final ethanol concentration of 70 %. Following 24 h of incubation at 4 °C, the precipitate was collected by centrifugation and air-dried in a fume hood to remove residual ethanol. The dried material was then redissolved in deionized water and dialyzed against distilled water for 72 h using a 3.5 kDa molecular weight cut-off (MWCO) dialysis membrane. Finally, the dialysate was concentrated, pre-frozen at −80 °C for ≥6 h, and lyophilized for 48 h to obtain crude PPP.

2.2.2. Determination of content

Quantitative analysis of phytochemical constituents was systematically conducted using validated spectrophotometric protocols. Total carbohydrate content was determined *via* the sulfuric acid-phenol method with glucose equivalents as the reference [24]. For flavonoid quantification, the aluminum nitrate-sodium nitrite colorimetric assay was performed using rutin standards for calibration [25]. Phenolic compounds were quantified relative to gallic acid using the Folin-Ciocalteu reaction kinetics [26]. Protein concentrations were measured using the Bradford method, where bovine serum albumin (BSA) as the calibration standard [27]. Finally, glucuronide levels were analyzed using the Blumenkrantz and Asboe-Hansen method, with galacturonic acid as the standard [28].

2.2.3. Monosaccharide composition

A modified 1-phenyl-3-methyl-5-pyrazolone (PMP) pre-column derivatization-high-performance liquid chromatography (HPLC) technique was used to assess the monosaccharide composition [29]. In brief, polysaccharide samples (5 mg) underwent acid hydrolysis in 2 mL trifluoroacetic acid (TFA) at 121 °C for 2 h. Following hydrolysis, the hydrolysates were dried under nitrogen gas, subjected to three methanol washes, and reconstituted in sterile water for downstream analysis. Concurrently, a gradient series of monosaccharide standards, including rhamnose (Rha), arabinose (Ara), galactose (Gal), glucose (Glc), xylose (Xyl), mannose (Man), galacturonic acid (GalA), glucuronic acid (GlcA), glucosamine hydrochloride (GlcN), aminogalactose hydrochloride (GalN), and fucose (Fuc), were prepared in serial dilutions and filtered.

For PMP derivatization, 200 μL aliquots were treated with 0.5 M NaOH and PMP-methanol solution, incubated at 70 °C for 1 h, followed by neutralization with 0.5 M HCl and chloroform extraction. The processed samples were then analyzed using a Thermo U3000 HPLC system equipped with a ZORBAX Eclipse XDB-C18 column (4.6 × 250 mm, 5 μm). Chromatographic separation employed isocratic elution (acetonitrile: phosphate buffer = 17:83, *v*/v) at 0.8 mL/min flow rate, with detection at 250 nm and column temperature maintained at 30 °C.

2.2.4. Molecular weight distribution

Molecular weight distribution was analyzed using high-performance gel permeation chromatography (HPGPC, LC-10 A, USA) equipped with BRT105–103-101 gel columns (8 × 300 mm) and a RID-20 A refractive detector [30]. The mobile phase was 1× PBS at a flow rate of 0.8 mL/min, with the column temperature maintained at 40 °C.

2.2.5. FT-IR spectroscopy

FT-IR spectra were recorded on a Nicolet 6700 spectrometer (Thermo Fisher, USA) in absorbance mode, scanning from 4000 to 800 cm^{-1} with a resolution of 4 cm^{-1} [31]. Three scans were recorded for each polysaccharide sample.

2.2.6. Surface morphology

Surface morphology was analyzed by scanning electron microscopy (SEM; FEI-Quanta FEG 250, Thermo Fisher, USA) at an accelerating voltage of 5.00 kV. Samples were mounted on aluminum stubs with conductive adhesive, sputter-coated with gold, and imaged at 100× and 1000× magnifications.

2.3. Preparation and characterization of NADES

2.3.1. Preparation of NADES

NADES were prepared by mixing hydrogen bond acceptors (HBAs) and donors (HBDs) in specific molar ratios (detailed compositions and corresponding ratios are provided in Table 1) [32]. The mixtures were heated to 80 °C with continuous stirring until homogeneous, stable, and transparent solutions formed. Residual water was subsequently removed under vacuum at 50 °C.

2.3.2. FT-IR spectroscopy

Six NADES samples were frozen at −80 °C for at least 24 h and freeze-dried for an additional 24 h to remove free water. The desiccated NADES were compressed into tablets, and their infrared spectra were obtained using FT-IR (400–4000 cm^{-1}) with three scans.

2.3.3. Polarity determination

Three dimensions made up the Kamlet-Taft parameter, a widely used comprehensive index for evaluating solvent properties: polarity/polarizability (π^*), hydrogen bonding acidity (α), and hydrogen bonding alkalinity (β). The solvent property parameters of the six NADES were ascertained in this investigation using three probes, Reichardt's ET(30), *N*, *N*-dimethyl-4-nitroaniline (NA), and 4-nitroaniline (DMNA) [33]. The trials were conducted in the manner described below: The 1×10^3 M methanol solution of the dye was created by mixing 0.2 mL of the dye solution with 2 mL of the generated NADES at room temperature. After that, vacuum concentration was applied to extract the methanol. The wavelength values of maximal absorption (λ_{max}) were recorded after scanning the absorption spectra in the 190–900 nm wavelength range using a UV spectrophotometer. Finally, the parameters were calculated from a previously published formula.

$$E_T(30) = \frac{28591}{\lambda_{max(\text{Reichar's dye}(30))}} \quad (1)$$

$$\beta = \frac{1.035\nu_{(DMNA)} + 2.64 - \nu_{(NA)}}{2.80} \quad (2)$$

$$\alpha = 0.0649E_T(30) - 2.03 - 0.72\pi^* \quad (3)$$

Where,

Table 1
The composition of NADES.

Number	HBA	HBD	HBA/HBD[a]	Abbreviation
1	L-Proline	Citric Acid	1:1	L-ProCA
2	L-Proline	Lactic Acid	1:1	L-proLA
3	L-Proline	Malic Acid	1:1	L-ProMA
4	Betaine	Malic Acid	1:1	BetMA
5	Betaine	Lactic Acid	1:1	BetLA
6	Betaine	Glycerin	2:1	BetGly

[a] HBA/HBD was given in molar ratios.

$$\nu_{(DMNA)} = \frac{1}{\lambda_{max(DMNA)} \times 10^{-4}} \quad (4)$$

$$\nu_{(NA)} = \frac{1}{\lambda_{max(NA)} \times 10^{-4}} \quad (5)$$

Here, $\lambda_{max(RD)\ [nm]}$ represented the maximum absorption wavelength of Reichardt's dye (30). $\lambda_{max(NA)\ [nm]}$ and $\lambda_{max(DMNA)[nm]}$ represented the maximum absorption wavelength of NA and DMNA.

2.3.4. Viscosity measurement

The viscosity of NADES was measured at 25 °C using a rotational viscometer (NDJ—5S, Shanghai Changji Geological Instrument Co., Ltd.) with a No. 1 rotor, operating at 60 rpm and torque percentages between 10 % and 90 % [34].

2.4. Solubility of PPP in various solvents

PPP solubility was measured following Sun et al.'s method with modifications [33]. Briefly, PPP was accurately weighed (10.00 mg ± 0.01 mg) and dissolved in various solvents to a final concentration of 1.00 g ± 0.10 mg, followed by vortex mixing. Subsequently, the mixtures were incubated in an oven at 60 °C for 12 h, 24 h, 48 h, and 72 h to determine saturation solubility. The saturated solutions were centrifuged at 15000 rpm for 10 min to remove insoluble residues, and the supernatants were collected.

The supernatant was diluted 100-fold with deionized water, and total carbohydrate content was quantified by measuring absorbance at 490 nm using the phenol-sulfuric acid method. Solubility (S) was calculated as:

$$S = (A_{490nm} - b)/a \times 0.9 \times 1 \quad (6)$$

where S is the solubility of PPP (g/100 g solvent), A_{490nm} is the absorbance at 490 nm, and a (slope) and b (intercept) are derived from the standard curve. The standard curve was obtained by dissolving 0.1, 0.2, 0.3, 0.4, and 0.5 g of glucose into 10 g of different NADES according to the above procedure.

The determination process was performed in triplicate. The experimental data were analyzed using SPSS *V*.17.0 ($p < 0.05$).

2.5. Fabrication and characterization of DPNMs

2.5.1. Fabrication of DPNMs

Initially, PPP was dissolved in NADES containing 30 % (*w*/w) water under continuous stirring for 48 h until reaching dissolution equilibrium [35]. Following this dissolution phase, the mixture was centrifuged (10,000 rpm, 20 min, 4 °C) to isolate the supernatant, designated as NADES-PPP. Subsequently, three NADES-PPP solutions (6 %, 8 %, and 12 % *w*/w) were prepared by vortex-mixing the supernatant with distilled water at defined mass ratios. To prepare the electrospinning precursor, PVA was incorporated into the system to achieve a final concentration of 8 % (*w*/w) [36]. The composite solution was homogenized in two stages: static hydration at 25 °C for 5 h, followed by magnetic stirring (800 rpm) for 8 h until optically transparent. After ensuring complete dissolution, residual air bubbles were eliminated through ultrasonic degassing (40 kHz, 30 min). The required PVA mass was calculated using the following formula:

$$m_{PVA} = \frac{C_{PVA} \times V_{total}}{1 - C_{ppp}} \quad (7)$$

where C_{pva} denoted the target polyvinyl alcohol concentration (8 %, *w*/w), V_{total} represents the total volume of the precursor solution, and C_{ppp} corresponds to the polysaccharide concentration within the NADES-PPP system.

The NADES-PPP-integrated nanofiber membranes (DPNMs) were fabricated *via* precision electrospinning. To commence, the precursor solution was loaded into the electrospinning system (Ucalery® ESN-300), with key parameters optimized as follows: A needle-collector distance of 15 cm, applied voltage of 25 kV (16 kV positive polarity, 9 kV negative polarity), 0.6 mL/h feed rate, and 36 ± 1 °C chamber temperature. Relative humidity was maintained at 45–55 % RH using climate control modules (baseline temperature: 25 °C).

Upon establishing stable electrospinning jets (confirmed by 0.8–1.2 μA current readings), membranes were deposited for precisely 2 h per A4-sized area to ensure uniform morphology. Concurrently, control membranes (NM) devoid of NADES-PPP were generated under identical operational conditions, substituting NADES-PPP with aqueous solvent equivalents. Following this phase, the resultant membranes were codified based on PPP content: DPNM-6 (6 % *w*/w), DPNM-8 (8 % w/w), and DPNM-12 (12 % w/w), thereby enabling systematic dose-response evaluation in downstream assays.

2.5.2. Morphology analysis

The morphology of DPNMs was analyzed by SEM (SU8010, Hitachi, Japan) [37]. Fiber diameters were measured from SEM images using ImageJ software. For each sample, 100 fibers were randomly selected to calculate the average diameter and statistical distribution.

2.5.3. FT-IR spectroscopy

A small piece of DPNMs was cut and pressed into a sheet, and the infrared spectra of different nanofiber membranes were recorded using FT-IR scanning in the range of 4000–450 cm^{-1}.

2.6. In vitro permeation studies

2.6.1. Fluorescent labeling of PPP

To begin the process, PPP (400 mg) was dissolved in 0.2 M phosphate buffer (pH 8.0) and incubated with L-tyrosine (Tyr) and sodium cyanoborohydride at 37 °C for 96 h [38]. After reaction completion, the supernatant was decanted. The precipitate was purified by dialysis (3.5 kDa MWCO) against distilled water for 48 h, followed by freeze-drying to obtain Tyr-PPP. Fluorescein isothiocyanate (FITC) was conjugated to Tyr-PPP through a 24 h reaction under dark conditions at pH 8.5 and 25 °C. The reaction mixture was precipitated with ethanol (final concentration 70 %), centrifuged at 10,000 rpm for 10 min, and the pellet was dialyzed (3.5 kDa MWCO) and lyophilized to isolate FITC-PPP. Fluorescence spectra (excitation/emission: 490/520 nm) were recorded to validate the covalent conjugation of FITC to PPP, with spectral characteristics consistent with successful labeling [39]. In the final stage, homogeneous FITC-PPP solutions were prepared, and nanoelectrospun membranes (FPDNMs) were fabricated *via* a calibrated electrospinning protocol.

2.6.2. Vertical franz diffusion cell experiment

To assess the transdermal performance, *in vitro* permeation studies were systematically conducted on DPNM-6, DPNM-8, DPNM-12, and NM (for blank control) nanofiber membranes. For these experiments, Franz diffusion were assembled using commercially procured, pretreated porcine skin to simulate physiological conditions [40]. Before testing, all membrane specimens were uniformly cut into circular discs (diameter: 1.2 cm) to match the diffusion cell's effective surface area (1.13 cm^2). Following this preparation, the skin substrate was rinsed with saline to ensure surface integrity and securely mounted between the donor and receptor chambers.

To standardize the permeation environment, pre-warmed PBS (pH 7.0) was introduced into the receptor compartment as a diffusion medium, with continuous agitation maintained at 100 rpm and a controlled temperature of 32 ± 0.5 °C. At predetermined intervals (2, 4, 6, 8, 10, 12, and 24 h), 1 mL aliquots were withdrawn from the receptor lumen and immediately replenished with fresh medium to sustain sink conditions. Three independent replicates of the permeation assay were

performed for each membrane type to ensure statistical robustness.

For quantitative analysis, the collected samples were subjected to the phenol-sulfuric acid method, a well-established spectrophotometric technique for total carbohydrate quantification. Subsequently, cumulative permeation per unit area was calculated using the following formula:

$$Q_m = \frac{C_n V + \sum_{i-1}^{n-1} V_i C_i}{A} \tag{8}$$

Where Q_m represents the cumulative permeability, while C_n and C_i denote the total sugar content of the sample points at the nth and i positions, respectively. V is the volume of the diffusion chamber, V_i is the volume of the sample liquid, and A is the permeable area (in cm^2).

Similarly, using the same experimental procedure and the pig skin previously mentioned, FITC fluorescent labeling was utilized to monitor the penetration of PPP into the skin. After FPDNM-6, FPDNM-8, FPDNM-12, and blank nanofibrous membranes NM without polysaccharides and NADES had penetrated for 24 h, the skins of the porcines were removed, cleaned with PBS, snap-frozen in liquid nitrogen, embedded, and ice-cut using the optimal cutting temperature (OCT). Following sample preparation (NM was labeled with DAPI), the penetration of polysaccharides in pig skin was observed using a fluorescence microscope for 24 h following fluorescence labeling.

2.7. Bioactivity analysis of DPNMs

2.7.1. Biocompatibility

The biocompatibility of DPNMs was evaluated using a CCK-8 assay with L929 fibroblast cells [41]. After being sheared into discs of 1.2 cm in diameter, the NM, DPNM-6, DPNM-8, and DPNM-12 nanofiber membranes were placed on an clean bench and sterilized by exposing their front and back surfaces to UV radiation for 2 h each. L929 cells were cultivated for 24 h at 37 °C in an incubator with 5 % CO_2 after being seeded in 96-well plates at a density of 4×10^3 cells/well. After cell attachment, the old medium was removed. Sterilized NM, DPNM-6, DPNM-8, and DPNM-12 membranes were then placed flat on the L929 cells, followed by the addition of 200 μL fresh drug-containing medium.

A negative control group (cells not receiving drug treatment), a blank control group (no cells, having DMEM media), and a drug-delivered group (containing NM, DPNM-6, DPNM-8, and DPNM-12) were all included in the experiment. For 24 h, treated 96-well plates were cultured in a cell culture incubator. Following a 24 h period, the 96-well plates were taken out, 10 μL of CCK-8 reagent was added to each well while protected from light, and the plates were put in a cell culture incubator for three hours before being taken out. A multifunctional enzyme marker was used to record the absorbance values at 450 nm, and formula (9) was used to determine the cell viability. With a minimum of three replicate wells in each group, the experiment was conducted three times.

$$Cell\ viability(\%) = \frac{A_{drug} - A_{blank}}{A_{negative} - A_{blank}} \tag{9}$$

$A_{negative}$ stands for the absorbance of the negative control group, A_{blank} for the absorbance of the blank group, and A_{drug} for the absorbance of the administered group.

2.7.2. Cell viability assay

The effect of nano-electrospun membrane samples on RAW264.7 cell viability was assessed using CCK-8 assay [42]. RAW264.7 cells were seeded at 5×10^3 cells/well in 96-well plates and incubated for 24 h. Sterilized nano-electrospun membrane samples (DPNM-6, DPNM-8, DPNM-12, NM) were co-cultured for an additional 24 h, followed by 10 μL CCK-8 solution addition and incubation for 3 h. Absorbance at 450 nm was recorded to determine cell viability, with all experiments performed in triplicate. The formula for cell activity was consistent with the previous formula (9) and will not be repeated here.

2.7.3. Nitrite production

The anti-inflammatory activity of DPNMs was measured *via* NO production using the Griess reaction [43]. After being adjusted to a density of 5.0×10^5 cells/mL, RAW264.7 cells in the logarithmic growth phase were seeded onto 96-well plates at a volume of 100 μL per well and cultivated for 24 h under standard conditions. Next, 50 μL of new medium containing test samples was added. The experimental groups included DPNMs (UV-irradiated nanofiber membranes DPNM-6, DPNM-8, and DPNM-12, cut into disc-shaped samples with an area of 1.13 cm^2), LPS (final concentration: 1 μg/mL) as the positive control, and an equal volume of full medium served as the blank control. The supernatant was collected after another 24 h of incubation, and a NO detection kit was used to measure the absorbance at 540 nm (OD_{540}) in order to quantify the release of nitric oxide (NO). The following formula was used to determine the ratio of OD values between the treated and blank groups in order to assess NO generation. Three independent replications of the experiment were conducted.

$$NO\ release\ rate(\%) = \frac{A_{druy} - A_{blank}}{A_{positive} - A_{blank}} \tag{10}$$

The experimental group was denoted by A_{druy}, the positive control group by $A_{positive}$, and the blank group by A_{blank}.

2.8. Antibacterial experiment

The antibacterial activity of DPNMs was evaluated using the colony counting method [41]. To initiate the bacterial culture preparation, frozen *S. aureus* stock was retrieved from −80 °C storage using a sterile inoculation loop, rapidly thawed, and aseptically streaked onto Luria-Bertani (LB) agar plates (designated as F1 plates). After a 24 h incubation at 37 °C, distinct single colonies from the F1 plates were selected and subcultured onto fresh LB agar plates (F2 plates) to ensure purity. Following this step, the F2 plates were incubated for an additional 24 h under identical conditions to obtain axenic bacterial cultures.

For subsequent experimental use, bacterial suspensions were prepared by homogenizing colonies from the F2 plates in sterile 0.85 % NaCl solution, which was adjusted to achieve a standardized concentration of 1×10^6 CFU/mL. Concurrently, UV-sterilized nanofiber membranes (NM, DPNM-6, DPNM-8, and DPNM-12) were trimmed into 5×10 cm segments and immersed in sterile tubes containing 5 mL of the prepared bacterial suspension. A membrane-free bacterial suspension was used as the blank control. All experimental and control groups were incubated at 37 °C for 16 h with continuous agitation at 100 rpm to maintain homogeneous interaction conditions.

Post-incubation, bacterial suspensions were serially diluted 1000-fold, and 50 μL aliquots of the diluted suspensions were uniformly spread onto fresh LB agar plates. To quantify viable bacteria, plates were incubated at 37 °C for 24 h, after which colonies were enumerated and photographed. Finally, the bactericidal efficiency of each membrane was calculated using the formula:

$$Bactericidal\ Rate(\%) = 1 - \frac{CFU_{experiment}}{CFU_{blank}} \tag{11}$$

where the numbers of plate colonies in the experimental group and the blank group are denoted by $CFU_{experiment}$ and CFU_{blank}, respectively.

The revival, subculture, and plating phases for *E. coli* were identical to those for *S. aureus*, while *P. acnes* needed Reinforced Clostridium Agar (RCA) plates and anaerobic incubation in anaerobic jars with oxygen-absorbing packs. All other techniques were the same.

2.9. Statistical analysis

All the experiments were carried out in technical and biological triplicates. Statistical analysis was conducted with GraphPad Prism 10.0 (GraphPad Software Inc., USA). All data are expressed as the mean ± SEM. The level of statistical significance was determined by *t*-test and one-way ANOVA or two-way ANOVA for multiple comparisons. The data were considered statistically significant when $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, and $^{****}p < 0.0001$.

3. Results and discussion

3.1. Structural characterization of PPP

3.1.1. PPP extraction analysis

In this study, PPP was isolated using ultrasonication-assisted alkaline extraction. Following alkaline extraction, ethanol precipitation, and lyophilization, the procedure produced 83.41 g of crude polysaccharides (41.71 % extraction yield) from 200 g of dry raw material. A refined yield of 14.25 % was obtained after further purification using lyophilization and dialysis.

3.1.2. Compositional analysis of PPP

Fig. 1A displayed the contents of total sugar (TS), uronic acid (UA), total phenolic acid (TPC), total flavonoid (TF), and protein (PRO) in PPP, with the total sugar content being 55.52 % ± 9.22 %. To further streamline the process and reduce costs while adhering to the principles of green chemistry by minimizing the use of chemical reagents, purification steps were omitted, and PPP were not subjected to further purification. Consequently, PPP retained a certain proportion of UA, PRO, and other residual components.

3.1.3. Analysis of monosaccharide composition of PPP

The neutral monosaccharide composition of PPP was determined by PMP and HPLC. The total ion chromatogram (TIC) (Fig. 1B) indicated that PPP mainly consisted of glucose, galactose, and arabinose at molar

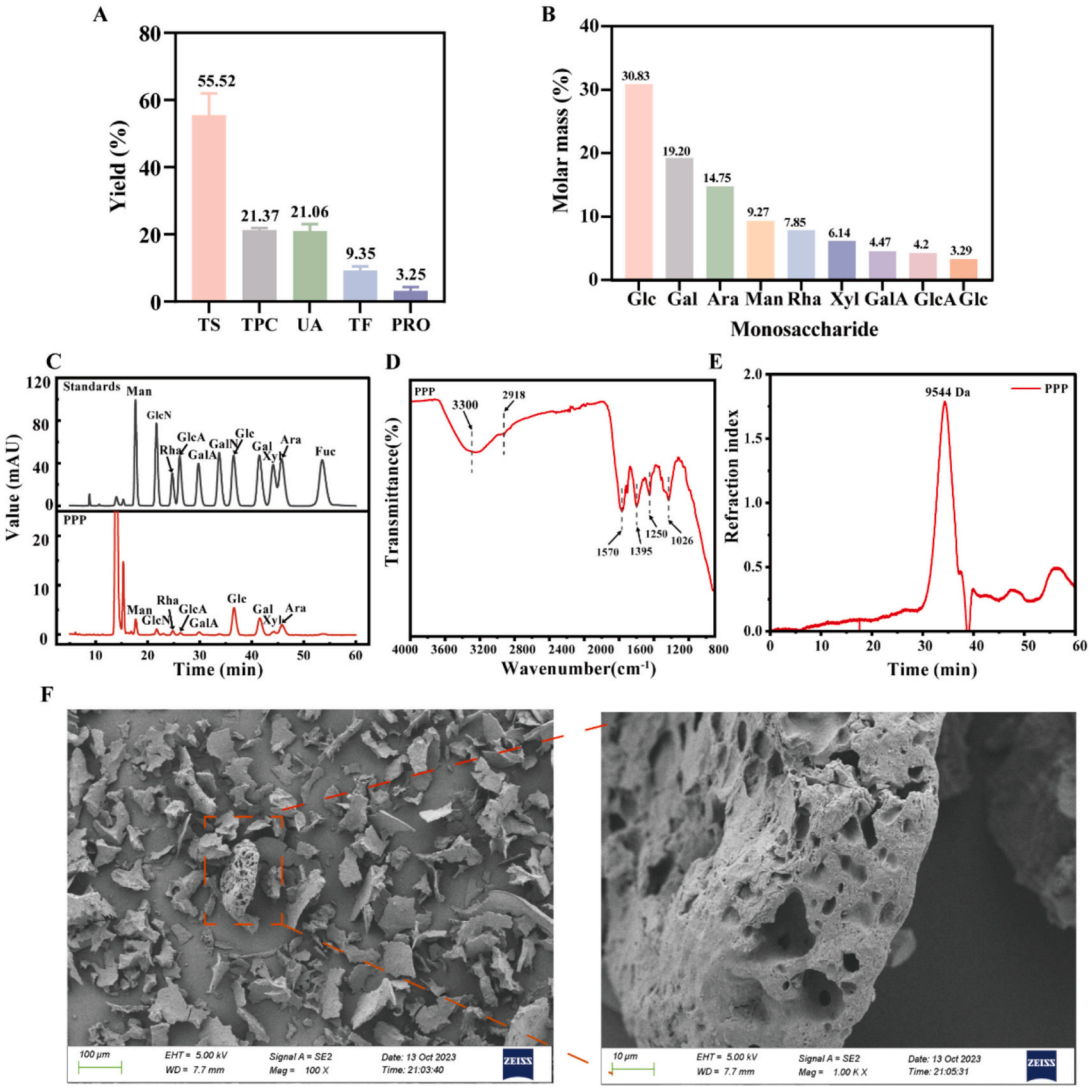


Fig. 1. (A) The contents of total sugar (TS), uronic acid (UA), total phenolic acid (TPC), total flavonoid (TF), and protein (PRO) in PPP were determined through colorimetric assays, with numerical values on the bar chart representing the mean percentage content of each component. (B) Schematic representation of the monosaccharide composition in PPP, with numerical labels on the bar chart denoting the mean mass percentage content of individual monosaccharide constituents, illustrating their relative proportions. (C) HPLC profiles of monosaccharide standards and PPPs, showing the elution patterns and peak assignments for individual monosaccharides. (D) FT-IR spectrum of PPP, displaying the characteristic absorption bands corresponding to its molecular structure. (E) The molecular weight distribution of PPP determined by the HPGPC method showed that the Mw of PPP was 9544 Da. (F) The SEM images of PPP were magnified at 100× and 1000× to demonstrate its morphological characteristics and structural features.

percentages of 30.83 %, 19.20 %, and 14.75 %, respectively (Fig. 1C), which was consistent with previous research findings [44].

3.1.4. Infrared spectral analysis of PPP

The Fourier-transform infrared spectrum of PPP was recorded in the range of 4000–500 cm^{-1} to elucidate its structural characteristics (Fig. 1D). The spectrum exhibited key absorption bands consistent with polysaccharide functional groups: a prominent broad peak at 3300 cm^{-1} corresponded to O—H stretching vibrations, while a weak signal at 2918 cm^{-1} arose from C—H stretching modes. Distinct asymmetric and symmetric stretching vibrations of carboxylate groups (COO^-) were observed at 1570 cm^{-1} and 1395 cm^{-1}, respectively. Additional features included a peak at 1250 cm^{-1}, likely attributed to pyranose ring vibrations or C—O stretching, and a strong absorption band at 1026 cm^{-1} that confirmed the presence of C–O–C glycosidic linkages characteristic of pyranose ring structures. These spectral signatures collectively validated the chemical identity of PPP as a polysaccharide, aligning with its molecular framework [45].

3.1.5. Molecular weight analysis of PPP

According to HPGPC (Fig. 1E) analysis, the average molecular weight of the main fraction of PPP was found to be 9544 Da. The molecular weight of PPP was lower than that of the polysaccharides extracted by Lihua Lin and others [23], which may be attributed to the ultrasonic-assisted extraction method employed in this study. The mechanical disruption and cavitation effects generated by ultrasonic waves likely result in the cleavage of polysaccharide molecular chains, thus reducing their molecular weight [46].

3.1.6. Surface morphology analysis of PPP

As shown in Fig. 1F, the microstructure of PPP exhibited an irregular flake-like morphology with curling features. At 1000× magnification, the surface appeared rough and cracked, with numerous pores observed within its internal structure. The images revealed that the surface of PPP had a high degree of roughness, likely resulting from the interactions and aggregation of polysaccharide chains. This rough texture may enhance the adsorption capacity of the polysaccharides. Additionally, the particles of PPP appeared irregular in shape and varied in size, possibly due to the random aggregation and crystallization behavior of polysaccharide molecules during extraction and drying. Furthermore, some porous structures observed in the images may be caused by the evaporation of water during the drying process, and these pores could affect the solubility and bioavailability of the polysaccharides [47].

In summary, a detailed analysis of PPP's structural properties and chemical composition was conducted. Its distinct microstructure, high total sugar content, and unique monosaccharide profile all demonstrated its potential for diverse industrial applications in a variety of industries. Future research will prioritize investigating PPP's functional characteristics and bioactivities.

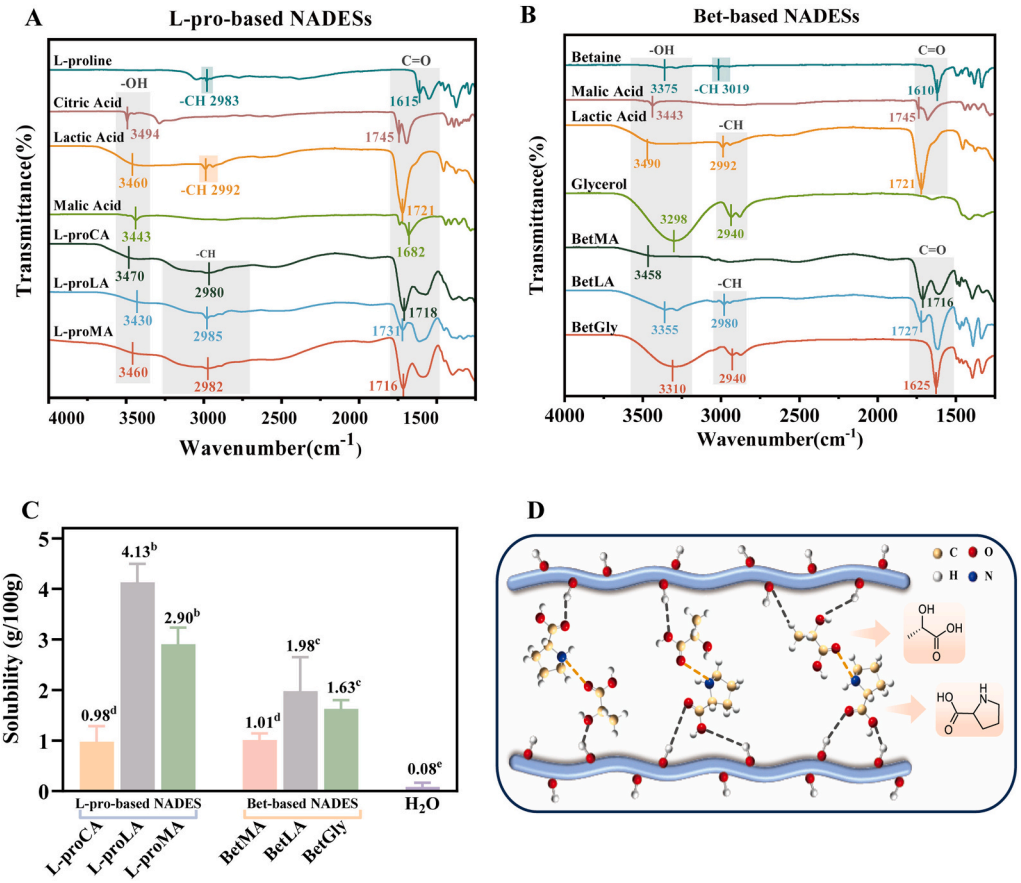


Fig. 2. (A) FT-IR spectrum of L-pro-based NADES, showing the characteristic absorption peaks associated with their molecular structures. (B) FT-IR spectrum of Bet-based NADES, highlighting the specific functional groups present in these solvents. (C) The schematic diagram illustrated the solubility of PPP in different NADES solvents, with H_2O as the control group. In the bar chart, higher numerical values indicated stronger PPP dissolution capacity. Letters above the bars denoted statistically significant differences between groups: identical letters (*e.g.*, 'a, a') indicated no significant difference, while distinct letters (*e.g.*, 'a, b') represented significant differences, with 'a' marking the most pronounced divergence. (D) Mechanistic diagram illustrating the interaction between polysaccharides and NADES, providing insights into the molecular interactions and possible hydrogen bonding formations.

3.2. Solubilization effect of NADES on PPP

3.2.1. Infrared spectral analysis of NADES

In this study, it was observed that PPP exhibited poor water solubility, posing challenges for its subsequent applications. Previous research has demonstrated that NADES can significantly enhance the dissolution of biomacromolecules through synergistic interactions between HBDs and HBAs [17,18]. Guided by this principle and in compliance with China's Inventory of Cosmetic Ingredients (China's IECIC), six NADES systems were designed and prepared using dermatologically approved components (Table 1) to explore their potential as biocompatible solvents.

For L-proline-based (L-pro-based) NADES (Fig. 2A), FTIR analysis showed that the –OH stretching vibrations in L-proCA, L-proLA, and L-proMA appeared at 3470 cm^{-1}, 3430 cm^{-1}, and 3460 cm^{-1}, respectively. These values differed notably from those of pure HBDs (citric acid: 3494 cm^{-1}; lactic acid: 3460 cm^{-1}; malic acid: 3443 cm^{-1}). Specifically, L-proCA and L-proLA exhibited red shifts of 24 cm^{-1} and 30 cm^{-1} in –OH peaks, while L-proMA displayed 17 cm^{-1} blue shift. This indicated that stronger O–H···O hydrogen bonds formed between L-proline and citric/lactic acids, reducing electron density in hydroxyl groups [48]. The anomalous blue shift in L-proMA suggested alternative intermolecular interaction patterns. Additionally, the carbonyl (C=O) vibration peak of L-proline at 1615 cm^{-1} blue-shifted to 1745 cm^{-1} (L-proCA), 1721 cm^{-1} (L-proLA), and 1682 cm^{-1} (L-proMA), signifying enhanced carbonyl polarization through N–H···O hydrogen bonding. Minor shifts and broadening in C—H vibrations (*e.g.*, L-proline's 2983 cm^{-1} peak shifting to 2980 cm^{-1} in L-proCA) further confirmed hydrogen-bond network reorganization and reduced molecular symmetry [49].

In Betaine-based (Bet-based) NADES (Fig. 2B), betaine's –OH vibration (3375 cm^{-1}) exhibited diverse shifts: a blue shift to 3458 cm^{-1} ($\Delta\nu$ = +83 cm^{-1}) in BetMA, red shifts to 3355 cm^{-1} ($\Delta\nu$ = −20 cm^{-1}) in BetLA, and 3310 cm^{-1} ($\Delta\nu$ = −65 cm^{-1}) in BetGly. The substantial red shift in BetGly implied strong O–H···O^- ionic hydrogen bonds between betaine's carboxylate group and glycerol's hydroxyls. Conversely, the blue shift in BetMA likely arose from unique dipole-dipole interactions or localized electronic environment alterations between betaine and malic acid [50]. Betaine's C=O vibration (1610 cm^{-1}) showed pronounced blue shifts: $\Delta\nu$ = +106 cm^{-1} in BetMA (1716 cm^{-1}), $\Delta\nu$ = +117 cm^{-1} in BetLA (1727 cm^{-1}), and $\Delta\nu$ = +15 cm^{-1} in BetGly (1625 cm^{-1}). These variations suggested that ionic hydrogen bonding significantly enhanced carbonyl polarization, with interaction strength depending on HBD type. The dramatic shift in BetLA correlated with stronger hydrogen bonds between betaine's carboxylate and lactic acid's hydroxyl groups. Furthermore, betaine's –CH vibration (3019 cm^{-1}) red-shifted to 2980 cm^{-1} ($\Delta\nu$ = −39 cm^{-1}) in BetLA and 2940 cm^{-1} ($\Delta\nu$ = −79 cm^{-1}) in BetGly, while disappearing in BetMA. These shifts and peak broadening reflected hydrogen-bond-induced symmetry reduction, particularly evident in BetGly due to glycerol's multi-hydroxyl structure promoting complex hydrogen-bond arrangements [51].

The results demonstrated distinct hydrogen-bonding patterns: O–H···O/N–H···O interactions dominated in L-pro-based systems, while O–H···O^- ionic bonds characterized Bet-based systems. Notably, the substantial –OH red shifts in L-proCA and L-proLA indicated densely packed hydrogen-bond networks. Such strong interactions could disrupt polysaccharide crystalline structures and enhance solvent-solute compatibility, thereby improving PPP solubility. These findings align with the NADES design principle emphasizing synergy between hydrogen-bond strength and solvent polarity, providing a theoretical foundation for developing natural, eco-friendly polysaccharide solvents.

3.2.2. Analysis of PPP solubility

Fig. 2C illustrated the solubility determination of PPP in aqueous and NADES systems using a modified established method [14]. Only 0.08 g/100 g of PPP was soluble in water; in contrast, PPP was substantially more soluble in several NADES solvents. With a solubility of 0.98 g/100 g, the NADES solvent L-proCA demonstrated the least capacity to dissolve PPP and was 11.25 times more soluble in water than PPP. The best solubilizing effect of PPP was demonstrated by the low eutectic solvent L-proLA, which was composed of L-proline and lactic acid. This solvent's solubility in water was 51.63 times greater than that of PPP. Consequently, L-proLA was also selected for further tests in this thesis since it had the best solubilizing effect on PPP. In general, L-pro-based NADES had a greater solubilizing effect on PPP than Bet-based NADES.

3.2.3. Viscosity analysis of NADES

Table 2 displayed the results of the rotational viscometer used in this thesis to determine the viscosity of NADES at room temperature. With L-ProCA having the highest viscosity (1348.00 Pa·s) and BetGly having the lowest (1.14 Pa·s), the L-pro-based NADES system had a higher viscosity than the Bet-based NADES system. L-proCA had a higher viscosity than the NADES system based on Betaine.

3.2.4. Kamlet-Taft parameters of NADES

The results were displayed in Table 2. The π-π electron jumps of the 4-nitroaniline molecule, the length of the anions' and cations' chains, and the substituent species and locations in the NADES all directly influenced the values of π^*. The results showed that the L-pro-based NADES system's α-value was comparatively higher than that of the Bet-based NADES system, where L-proLA's α-value was the highest, reaching 2.904, indicating that L-proLA had the strongest capacity to provide hydrogen bonding. The structure of the anion mostly determined the β-value; if the ions had a high hydrogen bonding acceptance capacity, the NADES system's hydrogen bonding acceptance capacity was strong [50]. BetLA had the greatest β-value of 2.937, suggesting that it had a high capacity to absorb hydrogen bonds. The β-value from the L-pro-based NADES system was comparatively lower than that of the Bet-based NADES system [51].

3.2.5. Mechanistic analysis of PPP solubilization by NADES

This study systematically investigated the regulatory role of hydrogen-bond networks in enhancing polysaccharide (PPP) dissolution through an in-depth analysis of hydrogen-bond interactions in NADES. Key parameters reflecting hydrogen-bond network properties (*e.g.*, viscosity, polarity) were measured alongside dissolution experiments, revealing that NADES significantly improved PPP dissolution efficiency compared to pure water (Fig. 2C). Among these, L-pro-based NADES outperformed Bet-based systems, with the L-proLA formulation achieving the highest enhancement (51.63-fold increase). Further analysis (Table 2) demonstrated that L-pro-based NADES exhibited higher viscosity and hydrogen-bond donor capacity (α values), attributed to their molecular structure: carboxyl and amino groups in L-proline form stronger intermolecular hydrogen bonds, creating a dense three-dimensional network that restricts molecular mobility and elevates viscosity [50]. Concurrently, the elevated α values indicated enhanced proton-donating ability, effectively disrupting intramolecular hydrogen bonds in PPP to facilitate dissolution. These findings collectively demonstrated that L-pro-based systems, by optimizing hydrogen-bond donor capacity and network stability, more efficiently disassemble polysaccharide hydrogen-bond networks, thereby achieving superior

Table 2
The viscosity and the Kamlet-Taft parameters of NADES.

NADES		Viscosity (Pa·s)	*EN T*	π^*	α	β
L-pro-based	L-ProCA	1348.00	1.135	0.173	2.258	0.787
	L-proLA	10.13	1.420	0.119	2.904	0.577
	L-ProMA	476.00	1.352	0.128	2.752	0.474
Bet-based	BetMA	27.70	1.401	0.128	2.854	0.460
	BetLA	2.78	1.440	0.126	0.459	2.937
	BetGly	1.14	1.416	0.135	0.385	2.907

solubilization performance compared to Bet-based NADES.

Existing studies indicated that the core mechanism by which NADES enhanced polysaccharide dissolution lies in their ability to disrupt intra- and intermolecular hydrogen bonds within polysaccharides through hydrogen-bonding interactions, thereby reducing crystallinity (Fig. 2D). For instance, HBAs (*e.g.*, Cl^-) interact with hydrogen atoms of polysaccharide hydroxyl groups, while HBDs (*e.g.*, urea) bind to oxygen atoms, collectively forming new hydrogen-bond networks that weaken the original rigid structure of polysaccharides [51]. Furthermore, NADES penetrate crystalline regions of polysaccharides, disrupting their ordered arrangements. Ultrasonication-assisted NADES treatment, for example, significantly reduces cellulose crystallinity and enhances dissolution [17]. Additionally, the incorporation of moderate water content (5 %–30 %) reduces NADES viscosity, increases polarity, and partially replaces hydrogen bonds between NADES and polysaccharides, thereby balancing dissolution efficiency and system stability [18].

In this study, we found that L-pro-based NADES, with stronger hydrogen bond networks due to carboxyl and amine groups of L-proline, exhibited superior solubility for PPP compared to Bet-based NADES. L-proLA showed the highest solubility, emphasizing the importance of

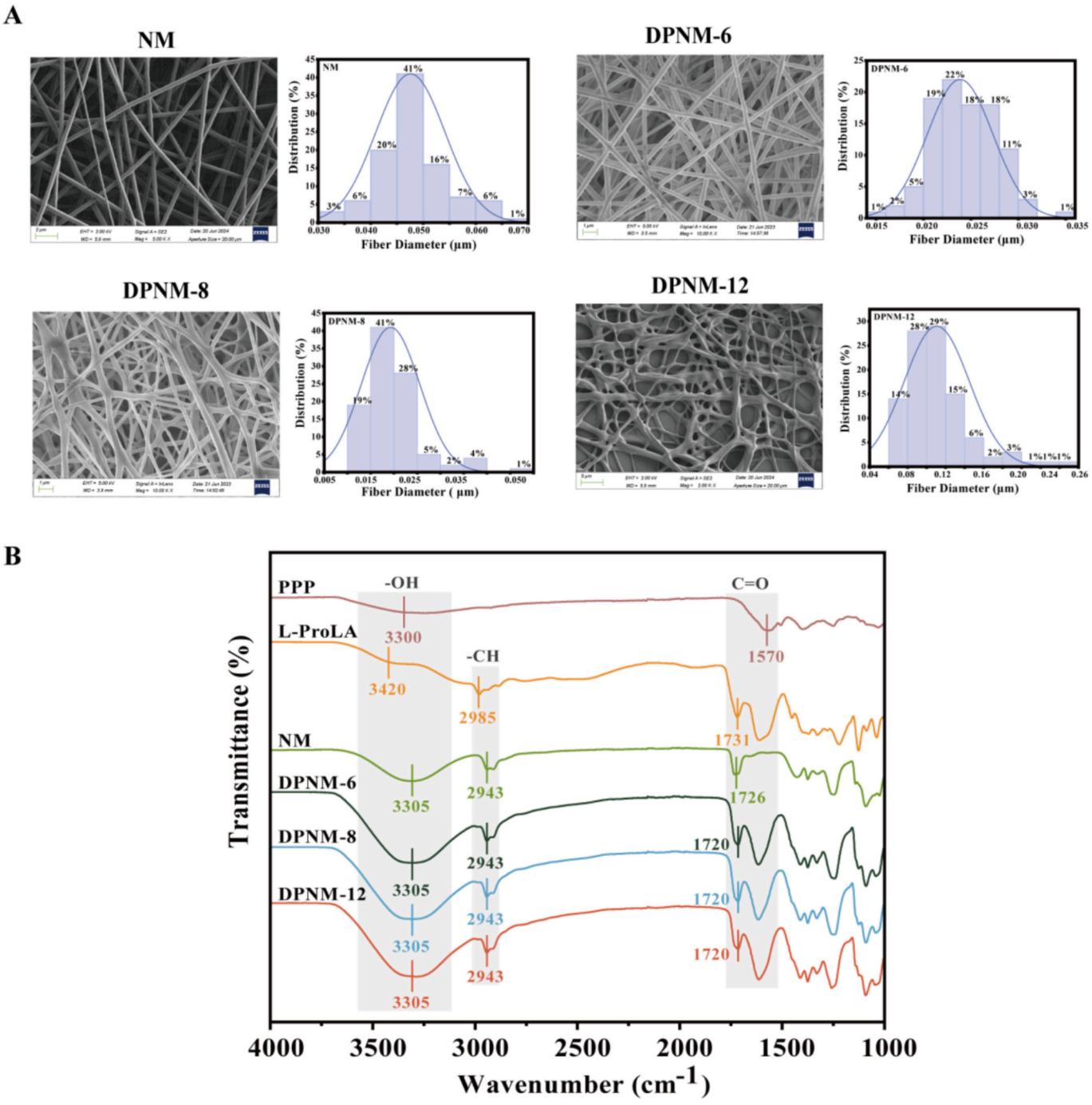


Fig. 3. (A) The SEM images of DPNMs dressings demonstrated the detailed morphology and structural features of the nanofibers, while the corresponding fiber diameter distribution histograms provided quantitative insights into fiber size and uniformity. Numerical labels on the distribution plots denoted the fiber count within specific diameter ranges, with higher values reflecting a greater abundance of fibers in those size intervals. (C) FT-IR spectrum of DPNMs, displaying the characteristic absorption bands that correspond to the molecular composition and functional groups present in the nanofiber membranes.

optimizing NADES composition for enhanced polysaccharide solubilization and broader industrial applications.

3.3. Fabrication and characterization of DPNMs

3.3.1. Preparation analysis of DPNMs

After the solubility issues in PPP were successfully resolved, this work implemented molecular-level design strategies to advance electrospinning technology through eco-friendly protocols. At 8 % (*w/v*), PVA, a water-soluble and biocompatible polymer, was used as biopolymeric matrix modifier to maximize molecular entanglement in the electrospinning solution. PVA's low toxicity and renewable nature meet sustainable material selection standards for green electrospinning [8]. Importantly, volatile organic compound emissions were reduced by substituting traditional organic solvents like chloroform and DMF [6].

3.3.2. Fiber structure analysis of DPNMs

SEM images and fiber diameter distributions of the electrospun membranes were shown in Fig. 3A. Solution viscosity is a critical parameter controlling fiber morphology: high viscosity (*e.g.*, DPNM-12) resulted in irregular fiber diameters from incomplete jet stretching, whereas low viscosity may cause droplet formation due to insufficient chain entanglement. On the other hand, uniform nanofibers were produced by moderate viscosity (DPNM-6), indicating the best possible balance between chain entanglement and electrostatic drawing. The SEM pictures showed that the NM fiber membrane without L-proLA-PPP had uniformly distributed diameters between 0.04 and 0.055 μm and tightly intertwined fibers that formed a mesh-like structure. Smaller average fiber diameters and a more uniform diameter distribution were the results of segmented liquid flow during spinning, which was made possible by the lower viscosity of the spinning solution without L-proLA-PPP. With fiber sizes mainly ranging from 0.02 to 0.03 μm, the DPNM-6 fiber membrane demonstrated straight, interconnected fibers creating a distinct network topology. It had a better shape and orientation, and it had smaller diameters than the NM fiber membrane.

According to these findings, a moderate concentration of L-proLA-PPP may improve the structural integrity of the nanofiber membrane network by increasing the spinning solution's viscosity, decreasing surface tension, lowering the average fiber diameter, and achieving a more uniform diameter distribution. Although a fibrous network was also generated by the DPNM-8 fiber membrane, the fibers showed signs of bending, which led to irregular fiber diameters and the development of some thicker individual fibers.

On the contrary, the network structure of the DPNM-12 fiber membrane showed unfavorable features, such as larger fiber diameters and an uneven distribution, with diameters mostly falling between 0.075 and 0.15 μm. The thicker and less homogeneous fibers were probably caused by the spinning solution's increased viscosity and surface tension, which prevented solution separation under electrostatic forces. The importance of surface tension and viscosity in attaining the ideal fiber distribution and morphology during electrospinning was shown by this analysis [52].

3.3.3. Infrared spectral characterization of DPNMs

The DPNM exhibited distinctive absorption peaks similar to those of the individual components, such as PVA, PPP, and the NADES (L-proLA), according to the FT-IR spectrum analysis illustrated in Fig. 3B. The distinctive hydroxyl peaks (approximately 3500–3300 cm^{-1}), C—H peaks (approximately 2900 cm^{-1}), and carbonyl peaks (approximately 1600–1700 cm^{-1}) found in PPP, PVA, and L-proLA are clearly visible in the DPNM, with corresponding shifts that confirmed the composition of the composite and confirming molecular interactions within it. Remarkably, NM's carbonyl peak was located at 1726 cm^{-1}, but DPNMs' was located at 1720 cm^{-1}. The redshift ($\Delta\nu = -6$ cm^{-1}) of the carbonyl peak in DPNMs was attributed to PPP incorporation of the absorption peak shifting to lower wavenumbers. The addition of polysaccharides encourages the material's hydrogen bonds to strengthen [53].

This study demonstrated the critical role of surface tension and viscosity in determining the shape of fibers during electrospinning. Variations in L-proLA-PPP concentration within PVA-based solutions produced nanofiber membranes with distinct properties. Excessive concentrations resulted in thicker, irregular fibers, whereas moderate concentrations increased viscosity, decreased surface tension, and improved fiber homogeneity. The integration of PVA, PPP, and L-proLA was highlighted by FT-IR analysis, which revealed hydrogen-bond formation between PPP hydroxyls and L-proLA carboxyl groups. These results provided important insights for biomedical and cosmetic applications by highlighting the necessity of balancing solution properties for the best nanofiber manufacturing.

3.4. The transdermal properties of DPNMs

3.4.1. Fluorescent labeling of PPP

Figs. 4A and B, respectively, demonstrated the FT-IR spectra of PPP and Typ-PPP, as well as the fluorescence spectra of FITC and FITC-PPP. The effective introduction of secondary amine groups *via* chemical modification to PPP, which resulted in the creation of the intermediate product Typ-PPP, was shown by the prominent absorption peak for Typ-PPP observed in the 3000–3500 cm^{-1} range. Additionally, the maximum absorption wavelength in the fluorescence spectra of FITC-PPP illustrated a blue shift when compared to FITC, indicating that FITC was not only adsorbed but also underwent nucleophilic addition with Typ-PPP, thereby labeling PPP [54].

3.4.2. In vitro transdermal analysis

The 24 h cumulative penetration kinetics of DPNMs nanofiber membranes in pig skin were presented in Fig. 4C. Although the L-proLA-PPP-free NM exhibited baseline penetration, the PVA matrix's UV–Vis spectrophotometric interference was probably the source of its signals, raising the possibility of detection artifacts. Superior penetration was demonstrated by the DPNM groups; at 12 h, the Q_m of the NM controls was 2.64 ± 0.30 mg/cm^2, whereas that of DPNM-6 (which had the lowest PPP content) was 5.49 ± 0.70 mg/cm^2. DPNM-6 had a negligible Q_m rise at 24 h (6.90 ± 0.20 mg/cm^2), suggesting that permeation was mostly present during the first 12 h. The highest PPP loading, DPNM-12, demonstrated a distinct dose-response relationship by achieving optimal permeation at 24 h (10.11 ± 0.13 mg/cm^2), followed by DPNM-8 (8.22 ± 0.20 mg/cm^2). According to these results, PPP permitted concentration-dependent transdermal administration in <24 h, and the loading content of PPP had a positive correlation with penetration efficiency.

After 24 h, FPDNMs in pig skin sections were shown in fluorescence scan pictures in Fig. 4D. The cross-sectional structure of the pig skin was stained by the control group using only DAPI. Green fluorescence from FITC appeared on the pig skin, indicating skin penetration. Distribution across the whole thickness of the skin was observed by fluorescence analysis, with FDPNM-12 exhibiting the strongest fluorescence intensity and the greatest accumulation close to the epidermis. By contrast, FDPNM-6 exhibited sparse fluorescence, whereas FDPNM-8 demonstrated moderate fluorescence. Greater penetration resulted from FDPNM-12's increased L-proLA-PPP concentration, with notable accumulation in surface layers and limited deeper penetration. These results confirmed that PPP successfully passed through epidermal barriers made possible by the combined action of cutting-edge nanotechnology and the NADES solvent system.

The results demonstrated that while PPP exhibited limited transdermal absorption efficiency due to its high molecular weight, its permeation performance was significantly enhanced by integrating it with NADES and fabricating nanofiber membranes *via* electrospinning technology. The underlying mechanism involved two synergistic effects: NADES improved polysaccharide solubility through hydrogen-bonding interactions, which facilitated the release of bioactive components,

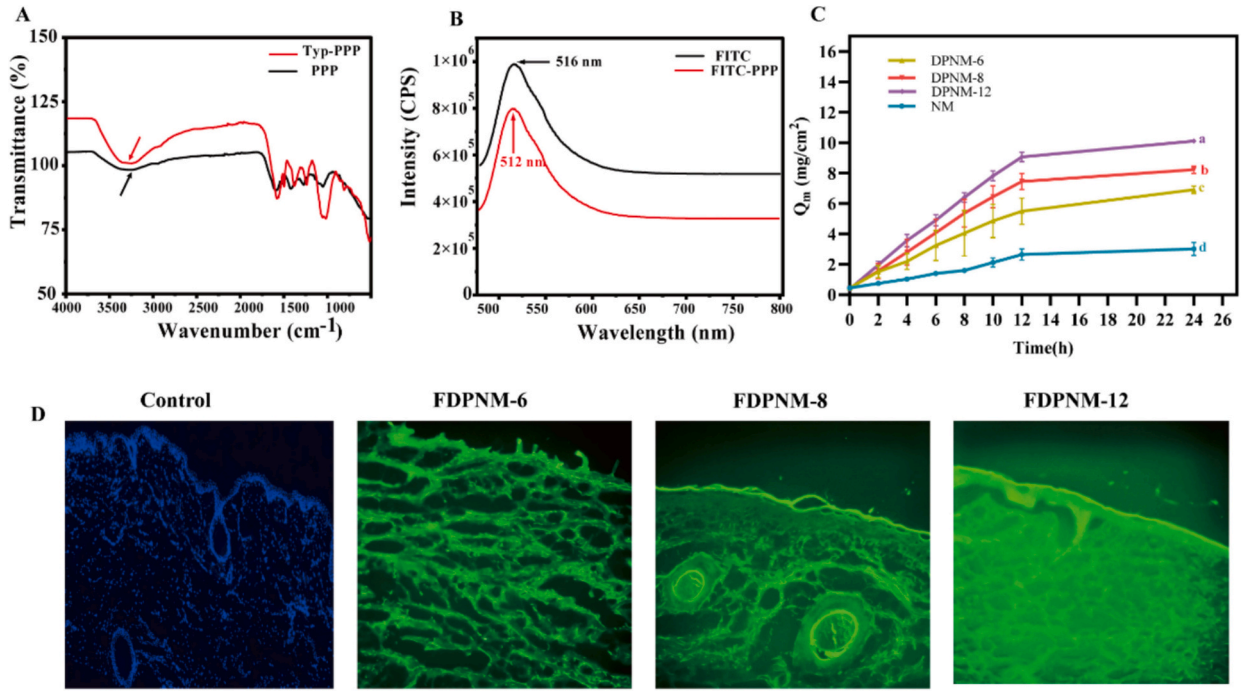


Fig. 4. (A) The comparative analysis of FT-IR spectra between PPP and Typ-PPP confirmed the successful incorporation of a secondary amine group into the PPP molecular structure, thereby verifying the formation of the intermediate product Typ-PPP. (B) Fluorescence spectral analysis revealed a blue shift in the maximum absorption wavelength of FITC-PPP compared to free FITC. This observation indicates that FITC was not merely physically adsorbed onto the surface of Typ-PPP but instead covalently conjugated *via* a nucleophilic addition reaction, thereby successfully achieving specific fluorescent labeling of PPP molecules. (C) Cumulative permeation profile of DPNMs nanofiber membranes on pig skin over a 24 h period demonstrates the time-dependent permeation behavior, with letters indicating statistically significant differences between groups (shared letters denote no significant difference; distinct letters indicate significant differences), where 'a' represents the highest significance level. (D) Fluorescence microscopy images of FDPNMs in cross-sections of pig skin after 24 h, highlighting the distribution and penetration depth of the nanofibers within the skin tissue. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

while lactic acid within NADES mildly acidified the stratum corneum, weakening its barrier properties and creating favorable conditions for transdermal delivery. Furthermore, encapsulating the NADES-PPP complex into electrospun nanofibers generated a nanostructured network that increased the effective contact area with the skin and enabled sustained release of PPP, synergistically boosting its permeation. This approach aligned with advanced strategies for optimizing bioactive compound delivery in dermatological applications [40].

Our study presented that PPP-loaded DPNMs nanofiber membranes effectively penetrate pig skin within 24 h in a dose-dependent manner, as higher L-proLA-PPP concentrations enhance cumulative permeation. Fluorescence analysis confirmed FITC-labeled PPP distribution throughout the skin, with DPNM-12 exhibiting the highest penetration and significant accumulation near the epidermis. These results highlighted the synergistic effects of the NADES solvent system and nanotechnology in overcoming skin barriers, offering valuable insights for advancing transdermal delivery systems and enhancing the bioavailability of polysaccharide-based formulations.

3.5. Bioactivity analysis of DPNMs

3.5.1. Biocompatibility of DPNMs

Biocompatibility serves as a critical prerequisite for dermal skin care formulations. The biocompatibility of DPNMs was evaluated *in vitro* using the CCK-8 assay to measure cell viability. As shown in Fig. 5A, statistical analysis revealed comparable cell viability across the control, NM, and DPNM groups, with NM and DPNMs exhibiting higher viability than the control. These results indicate that the electrospun nanofibers are cytocompatible and promote cell proliferation. The excellent biosafety profile (excellent biocompatibility and non-toxicity) of the nanofibers were further supported by the inherent biocompatibility of their components, including PVA, NADES, and plant polysaccharides, highlighting their potential application in skin care formulations.

3.5.2. In vitro anti-inflammatory activity of DPNMs

The biocompatibility of DPNMs with RAW264.7 macrophages was assessed using the CCK-8 assay. Results showed no significant cytotoxicity for both blank nanofiber membranes (NM) and drug-loaded DPNMs (Fig. 5B), confirming their suitability for subsequent anti-inflammatory evaluation. LPS, a specific activator of macrophages, was used to induce excessive release of inflammatory mediators such as nitric oxide (NO). The anti-inflammatory efficacy of DPNMs was evaluated by their ability to inhibit LPS-stimulated NO production (Fig. 5C). The NM group (92.5 ± 0.03 %) exhibited NO levels comparable to the LPS group (106.5 ± 0.04 %), indicating no inherent anti-inflammatory activity. In contrast, DPNMs demonstrated dose-dependent NO inhibition, with suppression rates increasing alongside L-proLA-PPP loading: DPNM-6 (48.9 ± 0.07 %), DPNM-8 (33.9 ± 0.02 %), and DPNM-12 (24.2 ± 0.03 %). These findings highlight the dose-dependent anti-inflammatory potential of DPNMs through active ingredient delivery, providing experimental evidence for developing anti-inflammatory materials based on polysaccharide-NADES composite systems.

3.5.3. Antibacterial activity of DPMNs

This study systematically evaluated the antibacterial properties of DPNMs using the plate counting method (Figs. 5F). Experimental data revealed that compared to blank nanomembranes (NM), DPNMs exhibited significant concentration-dependent antibacterial effects against *S. aureus*, *E. coli*, and *P. acnes* (Figs. 5D and E). At a low loading level (6 %, DPNM-6), the inhibition rates against *S. aureus*, *E. coli*, and *P. acnes* were 66.15 ± 1.21 %, 64.72 ± 0.82 %, and 36.01 ± 1.53 %, respectively. Notably, while both *S. aureus* and *P. acnes* are Gram-

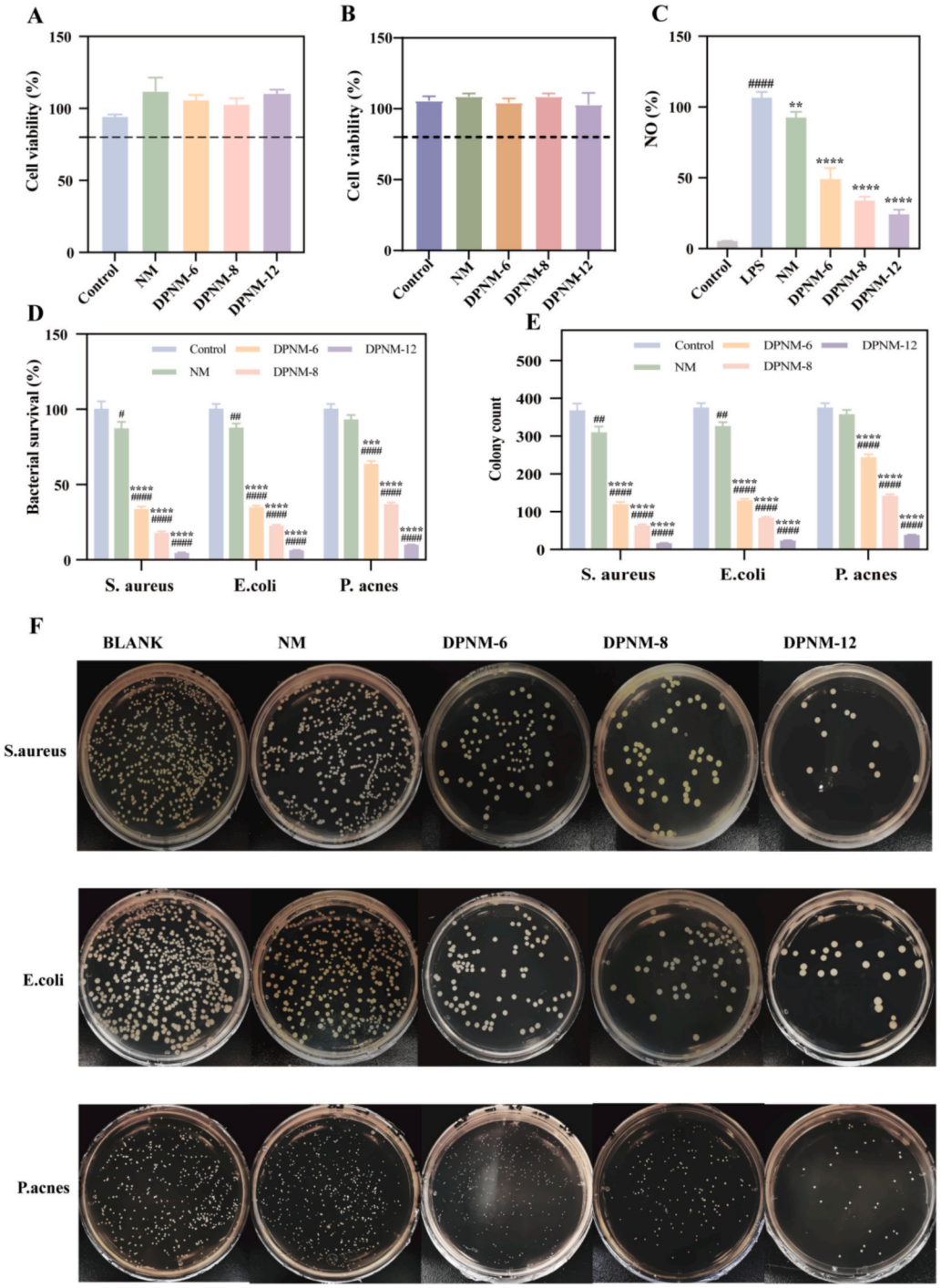


Fig. 5. (A) Biocompatibility of DPNMs determined by CCK8 method. (B) Toxicity profile of DPNMs on RAW264.7 cells. (C) Anti-inflammatory activity of DPNMs. Compared to the control, $^{\#\#\#}P < 0.001$. Compared to LPS, $^{*}P < 0.05$, $^{**}P < 0.01$, and $^{***}P < 0.001$. (D) Bacterial survival of *S. aureus*, *E. coli*, and *P. acnes*. Compared to control, $^{\#}P < 0.05$, $^{\#\#}P < 0.01$, and $^{\#\#\#}P < 0.001$. Compared to LPS, $^{*}P < 0.05$, $^{**}P < 0.01$, and $^{***}P < 0.001$. (E) Colony count of *S. aureus*, *E. coli*, and *P. acnes*. Compared to control, $^{\#}P < 0.05$, $^{\#\#}P < 0.01$, and $^{\#\#\#}P < 0.001$. Compared to control. $^{*}P < 0.05$, $^{**}P < 0.01$, and $^{***}P < 0.001$. (F) A comparative chart of the antimicrobial activity of nanomembranes containing varying concentrations of L-proLA-PPP against three different bacteria were presented, with the results obtained *via* the colony counting method.

positive bacteria, the markedly lower inhibition rate for *P. acnes* suggested that antibacterial activity might depend not only on bacterial classification but also on species-specific factors such as cell wall composition or metabolic traits. When the loading increased to 12 % (DPNM-12), the inhibition rates surged to 95.36 ± 0.17 % for *S. aureus*, 93.59 ± 0.15 % for *E. coli*, and 89.83 ± 0.24 % for *P. acnes*, demonstrating a strong positive correlation between L-proLA-PPP loading and antibacterial efficacy.

Further analysis integrating literature reports indicated that the superior inhibition of Gram-positive bacteria (*e.g.*, *S. aureus*) by DPNMs could be attributed to structural differences in bacterial cell walls. Specifically, the outer membrane lipopolysaccharide layer of Gram-negative bacteria (*e.g.*, *E. coli*) likely acts as a permeability barrier, limiting the penetration of NADESs components, whereas the thinner

peptidoglycan layer of Gram-positive bacteria facilitates easier access for antibacterial agents in DPNMs—a mechanism consistent with previous studies [55]. The low inhibitory effect on *P. acnes*, despite its Gram-positive classification, implies potential unique resistance mechanisms or metabolic adaptations in this species, warranting further investigation. Collectively, these findings not only confirm the pivotal role of L-proLA-PPP in the antibacterial functionality of DPNMs but also highlight their selective suppression of Gram-positive pathogens at lower loadings and broad-spectrum antimicrobial potential at higher concentrations. This work provides critical experimental insights for developing targeted anti-infective materials.

Overall, this study demonstrates the multifunctional bioactivity of DPNMs, including biocompatibility, anti-inflammatory, and antibacterial properties. CCK-8 assays confirmed cytocompatibility and cell proliferation enhancement, attributed to biocompatible components (PVA, NADES, plant polysaccharides). DPNMs showed dose-dependent anti-inflammatory effects by inhibiting NO production in LPS-stimulated RAW264.7 cells. Additionally, DPNMs effectively suppressed the growth of *S. aureus*, *E. coli*, and *P. acnes*, with higher L-proLA-PPP concentrations yielding greater antibacterial activity. These results highlight DPNMs' potential as advanced biomaterials for skin care.

4. Conclusion

This study successfully handled the solubility and transdermal problems of PPP by using NADES technology with electrospinning. L-proLA displayed extraordinary efficacy in dissolving PPP, with a 51.63-fold increase in solubility (4.13 g/100 g *vs.* 0.08 g/100 g in water) as compared to aqueous systems. Hydrogen-bond interactions between L-proline carboxyl and amino groups and PPP hydroxyl networks impair intramolecular bonds and diminish crystallinity, as indicated by FT-IR spectrum shifts and Kamlet-Taft parameter analysis ($\alpha = 2.904$). The addition of 30 % water improved viscosity and polarity, balancing dissolving efficiency with system stability.

The NADES-PPP/PVA system produced nanofibrous membranes with remarkable functionality (diameters ranging from 40 to 150 nm) using electrospinning. Because lactic acid weakens the stratum corneum barrier and the nanofibrous matrix promotes sustained release, DPNM-12, which contains 12 % *w*/w PPP, achieved 10.11 ± 0.13 mg/cm^2 cumulative skin permeation over 24 h in pig models. The system's ability to deliver bioactive substances under control was demonstrated by fluorescence imaging, which confirmed dose-dependent penetration and showed that PPP could pass through epidermal layers and accumulate in dermal areas.

In addition to >90 % fibroblast viability, the membranes demonstrated multifunctional efficacy, exhibiting 95.36 ± 0.17 % inhibition of *S. aureus*, 93.59 ± 0.15 % suppression of *E. coli*, and 75.8 % reduction in inflammatory nitric oxide release. These characteristics, along with the solvent-free manufacturing method and biodegradable ingredients, make DPNMs environmentally friendly options for skincare products that target acne by reducing inflammation and microbial growth.

This work promoted sustainable polysaccharide use in cosmetics by fusing nanotechnology and green chemistry concepts. Future work will concentrate on assessing NADES's recyclability, scaling up production procedures, and using *in vivo* validation to convert laboratory results into clinical efficacy.

CRediT authorship contribution statement

Wenfeng Li: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Bingjinfeng Zhan:** Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Zijing Peng:** Methodology, Investigation, Data curation, Conceptualization. **Yingchao Wang:** Writing – review & editing, Methodology, Conceptualization. **Zhongcan Peng:** Writing – review & editing, Supervision. **Yuanqiang Ling:** Methodology, Investigation. **Guomin Zhao:** Methodology, Investigation. **Depo Yang:** Writing – review & editing, Funding acquisition. **Zhimin Zhao:** Writing – review & editing, Supervision, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that there is no conflict of interest regarding the publication of this article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2025.144418.

Data availability

Data will be made available on request.

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